

Sayan Kumar Pattanayak

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EDUCATION:

Bachelor of Technology (B.Tech) - Computer Science & Engineering (AIML)
Galgotias University, Greater Noida | 8.67CGPA (2025-2029)

Class XII (CBSE, Science PCMB) Raisina Bengali School, Delhi | **70%** (Year of Passing: 2024)

Class X (CBSE) Raisina Bengali School, Delhi | **91%** (Year of Passing: 2022)

TECHNICAL EXPERTISE:

- **Programming Languages:** Python, C, C++
- **Web Development:** HTML5, CSS3
- **Data Science & Visualization:** NumPy, Pandas, Matplotlib, Seaborn
- **Frameworks & Libraries:** PyTorch, TensorFlow, Scikit-learn, OpenCV, Pandas, NumPy
- **Developer Tools & OS:** Git, GitHub, Linux (Ubuntu), Docker, VS Code, Solidworks

RESEARCH EXPERIENCE:

Independent Researcher – Computational Biology & AI
LLM-Based Cancer Gene Panel Validation using cBioPortal & TCGA
July 2026 – Present

- Designed and developed a research pipeline to evaluate **LLM-generated cancer gene panels** against real-world genomic datasets.

- Retrieved and analysed mutation and copy number alteration (CNA) data from **cBioPortal/TCGA** for multiple cancer types, including **Breast Cancer (BRCA)** and **Glioblastoma (GBM)**.
- Built automated data preprocessing, statistical analysis, and visualization workflows using **Python, Pandas, NumPy, and Matplotlib**.
- Benchmarked AI-generated gene panels against observed genomic mutation frequencies to assess their biological relevance.
- Developed reproducible, open-source research code with comprehensive documentation and data visualization.
- Collaborated with the **cBioPortal open-source community**, contributing improvements through merged pull requests and technical discussions.

Technologies: Python, Pandas, NumPy, Matplotlib, cBioPortal API, TCGA, Bioinformatics, Git, GitHub

GitHub: <https://github.com/sayan-aiml/llm-cancer-gene-panels>

CERTIFICATIONS & TRAINING

- **JPMorganChase - Software Engineering Job Simulation - Forge**

LINKS

- GitHub: <https://github.com/sayan-aiml>
- LinkedIn: <https://www.linkedin.com/in/sayan-kr-pattanayak>

I hereby declare that the above said particulars are true to the best of my knowledge and belief.

New Delhi
Sayan Kumar Pattanayak